Engineering Mechanics I

Week 5

Yuxia Yuan

Autonomous Aerial Systems

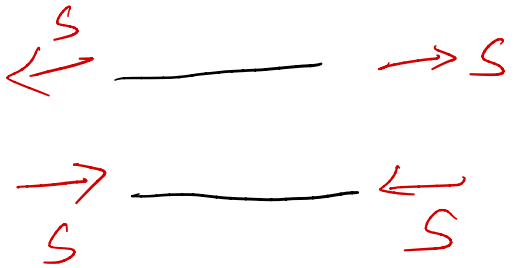
Trusses - Definition

Bars are

- straight
- have frictionless joints (pins/nodes) at both ends
- only take tensile or compressive forces in their axis, i.e., no moment effect

S > 0 → tensile

S < 0 → compressive



Trusses are

- structures of bars
- connected by pins / nodes
- external loads **are only applicable on the pins / nodes**

Trusses – Statically determined

Counting criterion in 2D says

$s + b \begin{bmatrix} = \\ < \\ > \end{bmatrix} 2n$

→ fulfilled
→ kinematic (movable)
→ statically overdetermined

→ even if CC fulfilled, it can be **movable.**

with

S → # **S**upport reactions
B → # **B**ars
N → # **N**odes

Kinematic determinacy

I → internally K. determined + externally K. determined

(if and only if) → PC

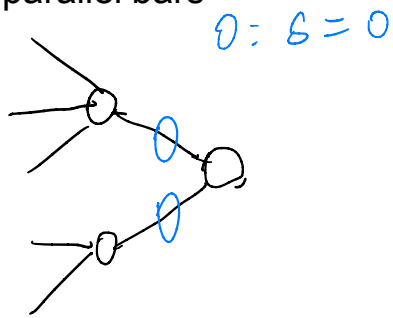
To calculate internal reaction by using static equilibrium equations, { Contiguity Criterion / Kinematic Det } need to be fulfilled

Trusses – Zero force bar

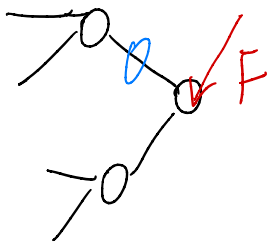
→ Bar with **no acting forces**

- Can be identified with the following rules of thumb:

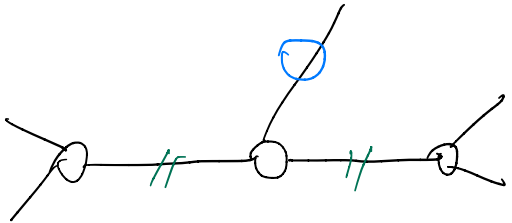
(I) Unloaded node with 2 non-parallel bars



(II) Loaded node with 2 bars and force in the direction of the bar



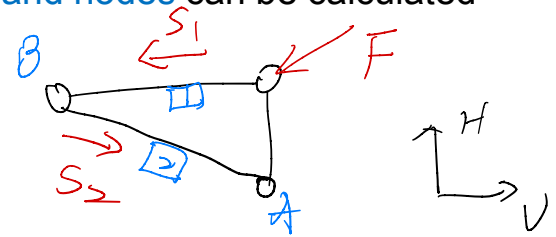
(III) Unloaded node with 3 bars, 2 of which are collinear



- Notice that:
  - it is a iterative process if there are changes in the external forces
  - zero bars cannot be removed →

Trusses – Computational Method of Joints

- Cutting principle at each node results in 2 equilibrium equations
- If truss is statically determined, forces acting in all bars and nodes can be calculated
- Can lead to a very big system of equations
- Directly applicable if there only are two unknowns
- Applicable in the vicinity of known disturbances (rule of thumb)



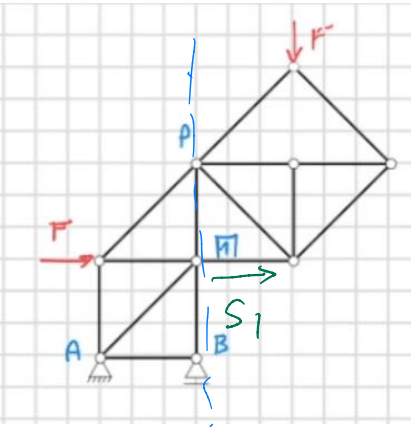
A: $\Sigma F_H = 0 \Rightarrow S_1$

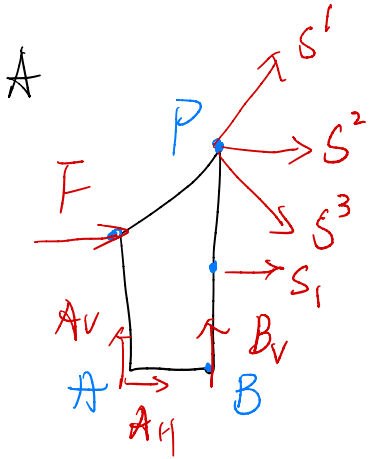
B: $\Sigma F_V = 0 \Rightarrow S_2$

Trusses – Method of Section / Cuts

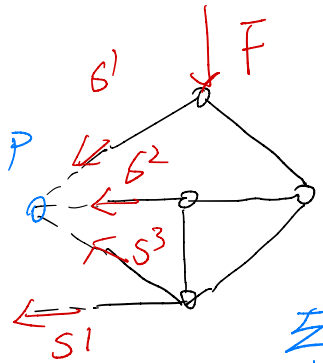
- Allows to calculate individual force members with only one equation
- Requires a “smart cut” to formulate the moment of equilibria around cleverly chosen points

Example:





+



$\sum M_P = 0$

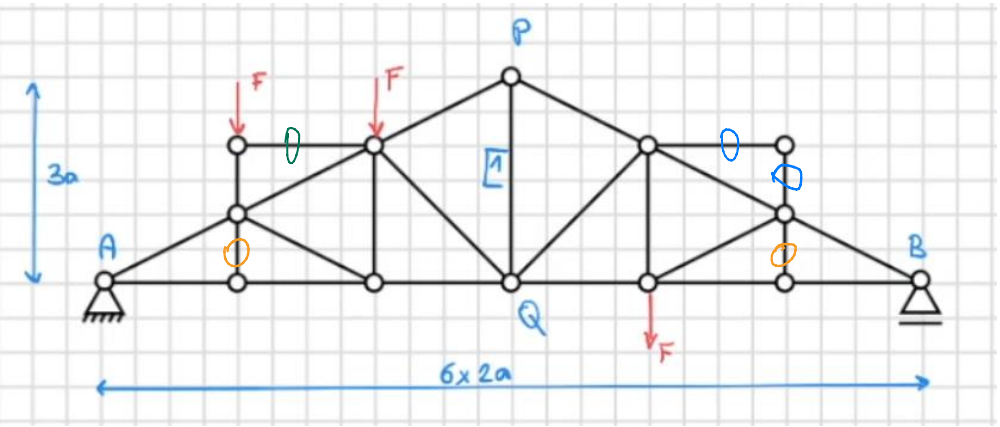
options:

1> statics in A

Exercise 1

We know: *Bar distance "a" and external force "F"*
Calculate: *Counting criterion (I), zero forces bars (II) and bar force S1 (III)*

Autonomous Aerial Systems



① Counting criterion

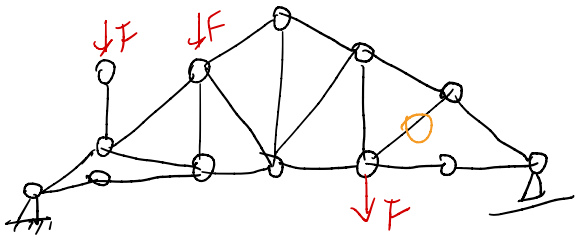
CC: $s+b=2$, $s=3$, $b=25$

$s+b=2\cdot 14$ ✓

→ { PC ✓ ; support: ✓ } ⇒ statically determined

② zero forces bars

1: 2
2: 1
3: 2+1
} b



③ Bar force S1

steps:
① static equilibrium { $\Sigma F=0$; $\Sigma M=0$ }
② Method of cuts:
③ Method of joints

Exercise 1

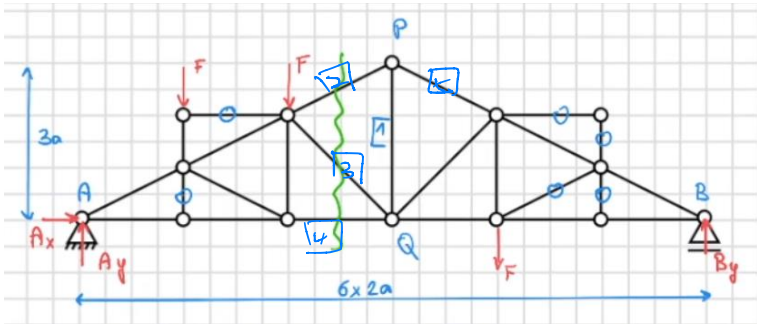
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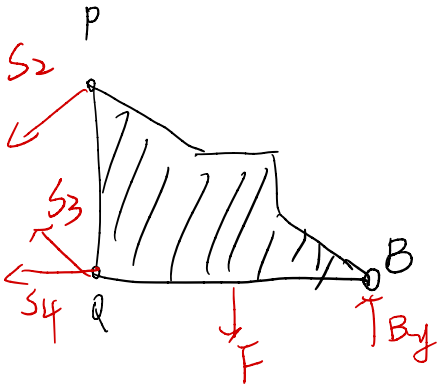
① $\Sigma F_x = 0,\ \Sigma F_y = 0,\ \Sigma M_A = 0.$

$$\begin{cases} \Sigma F_x = 0 \Rightarrow A_x = 0 \\ \Sigma F_y = 0 \Rightarrow A_y + B_y - 3F = 0, \\ + \circlearrowright \Sigma M_A = 0 \Rightarrow 2a\cdot F + 4a\cdot F + 8a\cdot F - 12a\cdot B_y = 0 \end{cases}$$

$$\Rightarrow \begin{cases} A_x = 0 \\ A_y = \frac{11}{6}F \\ B_y = \frac{7}{6}F \end{cases}$$



② Method of cuts



$+\circlearrowleft \Sigma M_Q = 0$

$3a\cdot S_{2x} - 2a\cdot F + 6a\cdot B_y = 0$

$S_2\cdot\cos\alpha = \frac{2\sqrt{5}}{5}$

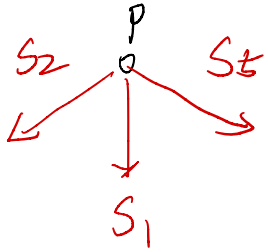
$S_2 = (-6B_y + 2F)\frac{\sqrt{5}}{6} = -\frac{5\sqrt{5}}{6}F$

Exercise 1

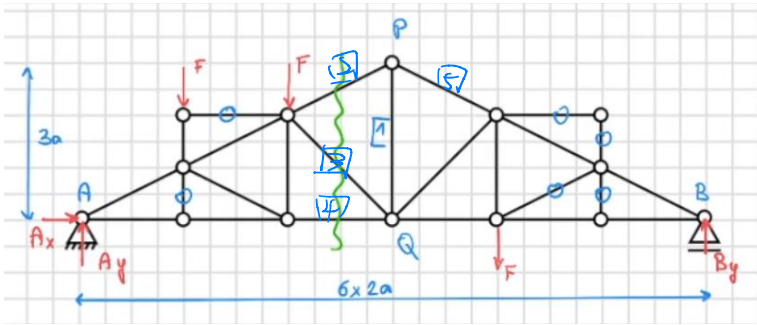
We know: *Bar distance "a" and external force "F"*
Calculate: *Counting criterion (I), zero forces bars (II) and bar force S1 (III)*

Autonomous Aerial Systems

③ Method of joints at P



two unknown



$\Sigma F_x = 0 \Rightarrow -S_{2x} + S_{5x} = 0 \Rightarrow -S_2 \cdot \cos\alpha + S_5 \cos\alpha = 0 \Rightarrow S_2 = S_5$

$\Sigma F_y = 0 \Rightarrow S_{2y} + S_{5y} + S_1 = 0 \Rightarrow S_1 = -(S_{2y} + S_{5y}) = -2S_2 \cdot \sin\alpha$

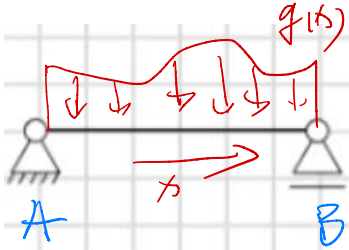
$= \frac{5}{3}F$

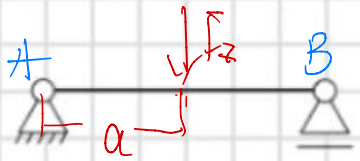
Continuous loads and resultants - Definition

Beams

~~Bars~~ are

Resultant force: $F_z = \int_A^B g(x)\,dx$



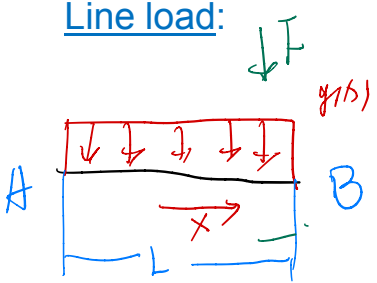


Application location: x_A

Resultant moment: $M_{Fz} = F_z \cdot a = \int_A^B x\,dF_z = \int_A^B x\,g(x)\,dx \Rightarrow a = \frac{1}{F_z}\int_A^B x\,g(x)\,dx$

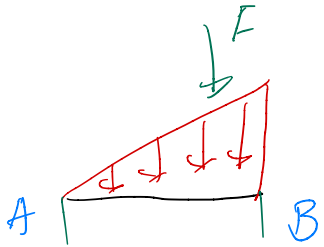
Examples:

Line load:



$F = g \cdot L$

$a = \frac{L}{2}$



External load:

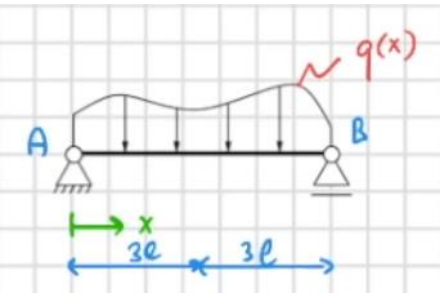
$F = \frac{gL}{2}$

$a = \frac{2}{3}L$

a: distance of the resultant force from the origin

Exercise 2

We know: *Beam distance "l" and external continuous load "q(x)"*
Calculate: *Resultant force(l) and application point "xA"(ll)*



$$q(x) = \begin{cases} q_0 \dfrac{x^2}{(3\ell)^2} & \text{für } 0 \le x \le 3\ell \\ q_0 \dfrac{(x-6\ell)^2}{(3\ell)^2} & \text{für } 3\ell < x \le 6 \end{cases}$$

$$F = \int_0^{6l} q(x)\,dx = \int_0^{3l} q_0 \frac{x^2}{(3l)^2}\,dx + \int_{3l}^{6l} q_0 \frac{(x-6l)^2}{(3l)^2}\,dx = 2q_0 l$$

$$a = \frac{1}{F}\int_0^{6l} x \cdot q(x)\,dx = \frac{1}{F} \cdot 6q_0 l^2 = 3l$$

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